Document 2: Fundamentals of Neural Networks

1. Architecture
Artificial Neural Networks consist of layers of interconnected nodes. Each connection has a weight that adjusts during training.
- Input Layer: Accepts raw features.
- Hidden Layers: Extract hierarchical patterns and non-linear representations.
- Output Layer: Produces final predictions or classification logits.

2. Training Mechanics
Optimization relies on forward propagation, loss calculation via a criterion function,
and backpropagation using gradient descent to minimize error across epochs.